
Exercise 1

A force $\vec{F} = 3\vec{\imath} + 4\vec{\jmath}\ (N)$ acts on an object as it moves from the origin to a final position M having coordinates x = 5.00 m and y=2.00 m. Start with the general definition for work of a force to find the work W done on the object by this force.

Exercise 2

Two blocks are connected by a very light string passing over a massless and frictionless pulley (Fig. 1). Traveling at constant speed, the 20.0 N block moves 72.0 cm to the right and the 12.0 N block moves 72.0 cm downward. How much work is done (a) on the 12.0 N block by (i) gravity and (ii) the tension in the string? (b) How much work is done on the 20.0 N block by (i) gravity, (ii) the tension in the string, (iii) friction, and (iv) the normal force? (c) Find the total work done on each block.

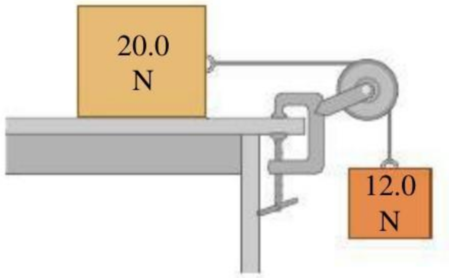
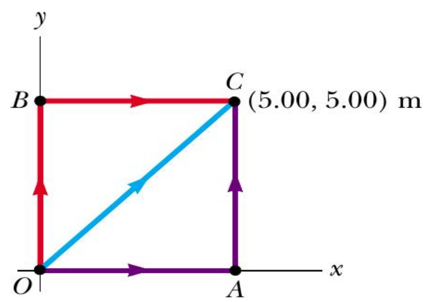


Figure 1

Exercise 3

Suppose that a constant force acts on an object. The force does not vary with time, nor with the position or the velocity of the object. Start with the general definition for work done by a force.

1. As a special case, suppose that the force $\vec{F} = 2\vec{\imath} + 3\vec{\jmath}\ (N)$ acts on a particle that moves from O to C (Figure). Calculate the work done by $\vec{F}$ if the particle moves along each one of the three paths OAC, OBC, and OC (Figure). (Your three answers should be identical.). What do you conclude?
2. Consider now the force $\vec{F} = 2y\vec{\imath} + x^2\vec{\jmath}\ (N)$, where x and y are in meters.
   a. Calculate the work done by $\vec{F}$ along (a) OAC, (b) OBC, (c) OC.
   b. Is $\vec{F}$ conservative or non-conservative?



Exercise 4

A 1.50 kg book is sliding along a rough horizontal surface. At point A it is moving at 3.21 m/s, and at point B it has slowed to 1.25 m/s. (a) How much work was done on the book between A and B? (b) If −0.750 J of work is done on the book from B to C, how fast is it moving at point C? (c)How fast would it be moving at C if +0.750J of work was done on it from B to C?

Exercise 5

Use the work–energy theorem to solve each of these problems. You can use Newton's laws to check your answers.

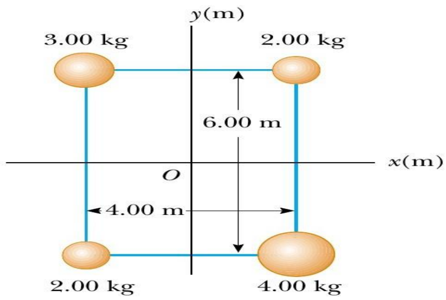
1. A skier moving at 5.00 m/s encounters a long, rough horizontal patch of snow having a coefficient of kinetic friction of 0.220 with her skis. How far does she travel on this patch before stopping?
2. Suppose the rough patch in part (1) was only 2.90 m long. How fast would the skier be moving when she reached the end of the patch?
3. At the base of a frictionless icy hill that rises at 25.0° above the horizontal, a toboggan has a speed of 12.0 m/s toward the hill. How high vertically above the base will it go before stopping?

Exercise 6

You are a member of an Alpine Rescue Team. You must project a box of supplies up an incline of constant slope angle α so that it reaches a stranded skier who is a vertical distance h above the bottom of the incline. The incline is slippery, but there is some friction present, with kinetic friction coefficient μ_k. Use the work-energy theorem to calculate the minimum speed you must give the box at the bottom of the incline so that it will reach the skier. Express your answer in terms of g, h, μ_k, and α.

Exercise 7

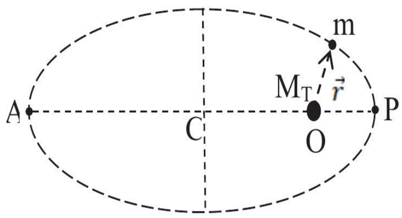
The four particles in Figure 4 are connected by rigid rods of negligible mass. The origin is at the center of the rectangle. If the system rotates in the xy plane about the z axis with an angular speed of 6.00 rad/s, calculate (a) the moment of inertia of the system about the z axis and (b) the rotational kinetic energy of the system



Exercise 8

A satellite of mass m = 1000 Kg has an elliptical trajectory about the Earth of mass M_T (Figure). Let r be the distance between the center O of the Earth and the satellite.

1. Show the forces exerted by the Earth on the satellite at point A and P. Show the velocity vector of the satellite $\vec{V}_A$ and $\vec{V}_B$.



2. The net mechanical energy of the satellite corresponds to

$$E_T = \frac{GM_T m}{2a}, \quad a = \frac{AP}{2}$$

G being the gravitational constant.

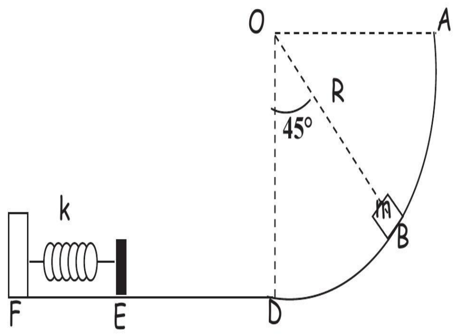
a. Give the satellite potential energy expression E_P in terms of G, M_T and r ($E_p(\infty) = 0$).

b. Find the satellite kinetic energy E_k in terms of G, M_T, a and r. Deduce the satellite velocity.

c. Calculate the satellite velocity at point A where OA = 41400 km, OP = 6750 km and $GM_T = 4.022 \times 10^{14}$ m³/s²

Exercise 9

In the vertical plane a mobile path is composed of quarter circle AB and a horizontal Part DEF. At point E lays a horizontal spring of constant k. The point- like mobile of mass m, is released from rest at point A.



1. If friction is neglected, find the mobile velocity at point B, the middle of AD .
2. Show the exerted force (1 N → 2 cm).
3. Find the acceleration a_B.
4. Find the maximum compression of the spring.

Numerical values: R = 1 m, m = 0.2 kg and k = 10^4 N/m